

Introduction to agents

Weekend 6 · More Agentic AI

Carlos Cotrini

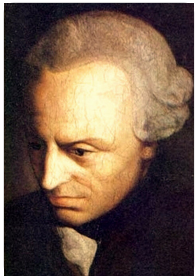


Section 1

Where we are

1

All the interest of my reason



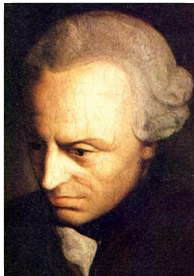
Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

- 1.
- 2.
- 3.

Kritik der reinen Vernunft, A805 / B833

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

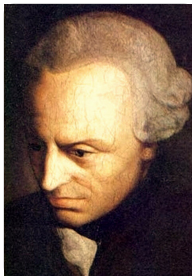
Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische) vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
- 2.
- 3.

What can I know?

Kritik der reinen Vernunft, A805 / B833

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

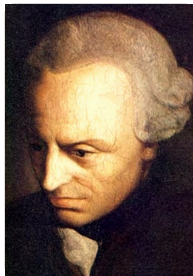
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1. Was kann **ich** wissen?
2. Was soll **ich** tun?
- 3.

What can I know? What ought I to do?

Kritik der reinen Vernunft, A805 / B833

All the interest of my reason



Immanuel Kant (c. 1790),
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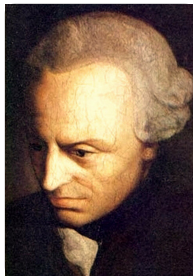
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1. Was kann **ich** wissen?
2. Was soll **ich** tun?
3. Was darf **ich** hoffen?

What can I know? What ought I to do? What may I hope?

Kritik der reinen Vernunft, A805 / B833

All the interest of my reason



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Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
2. Was soll **ich** tun?
3. Was darf **ich** hoffen?

What can I know? What ought I to do? What may I hope?

Kritik der reinen Vernunft, A805 / B833

Kant asked these of himself. We will ask them of the machine.

The same three questions, asked of the machine

Was kann *es* wissen? → Weekend 5: RAG

Was soll *es* tun? → Weekend 6: Agentic AI

Was darf *es* hoffen? → Weekend 7: Large Scale AI

Weekend 5 was what it can know. This weekend is what it should do.

Two days

Friday

- Agents, RL, and MCTS
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

Two days

Friday

- Agents, RL, and MCTS
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

Saturday

- Agent Q, and the limits of agentic AI
- Evolutionary coding agent
- Project

Two days

Friday

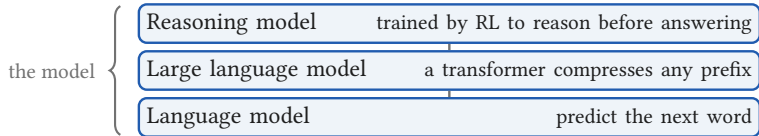
- Agents, RL, and MCTS ← we are here
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

This block is two half hours. The first is this lecture; the second is Monte Carlo tree search.

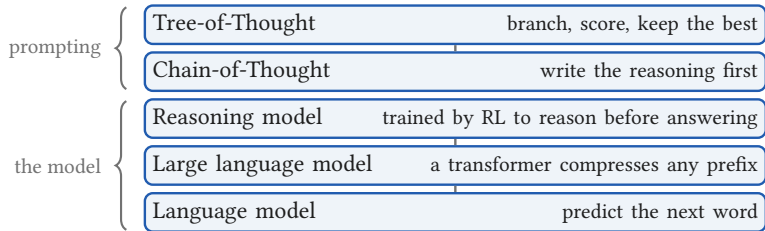
Saturday

- Agent Q, and the limits of agentic AI
- Evolutionary coding agent
- Project

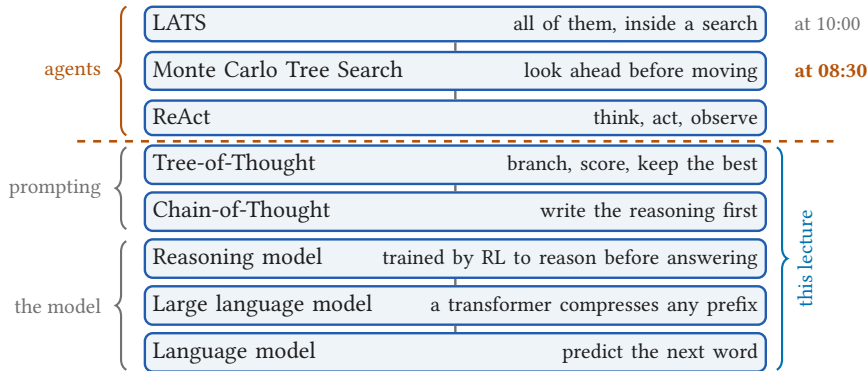
From a table to an agent



From a table to an agent



From a table to an agent

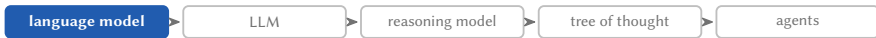




Section 2

Language models

2



A language model is a table

Next word					
coffee	tea	is	hot	cold	⟨end⟩

Columns: the vocabulary, every word the model can say

A language model is a table

Prefix so far

Next word

	coffee	tea	is	hot	cold	⟨end⟩
<i>nothing yet</i>						
coffee						
tea						
coffee is						
tea is						
coffee is hot						
⋮						

Rows: the prefix, everything said so far

A language model is a table

Prefix so far	Next word					
	coffee	tea	is	hot	cold	⟨end⟩
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Entries: how likely each word is to come next

A language model is a table

Prefix so far

Next word

	coffee	tea	is	hot	cold	⟨end⟩
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

$$\Pr(\text{hot} \mid \text{coffee is}) = 0.8$$

A language model is a table

Prefix so far

Next word

	coffee	tea	is	hot	cold	⟨end⟩
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

sums to 1

$$\Pr(\text{hot} \mid \text{coffee is}) = 0.8$$

Generating text is walking the table

Prefix so far

Next word

	coffee	tea	is	hot	cold	<end>
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

► read this row

Text so far

Generating text is walking the table

Prefix so far

Next word

	coffee	tea	is	hot	cold	<end>
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

► read this row

Text so far

coffee

Generating text is walking the table

Prefix so far

Next word

	coffee	tea	is	hot	cold	<end>
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
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coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

► read this row

Text so far

coffee

is

Generating text is walking the table

Prefix so far

Next word

	coffee	tea	is	hot	cold	<end>
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
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coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

► read this row

Text so far

coffee

is

hot

Generating text is walking the table

Prefix so far

Next word

	coffee	tea	is	hot	cold	<end>
<i>nothing yet</i>	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

► read this row

Text so far

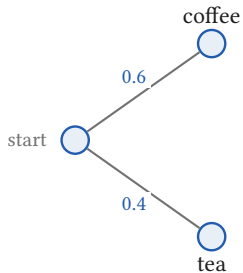
coffee

is

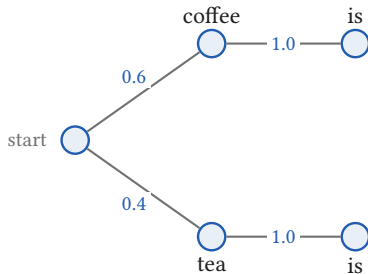
hot

and the model stops

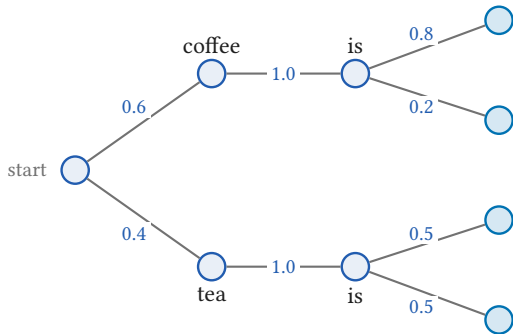
One table, every text it can write



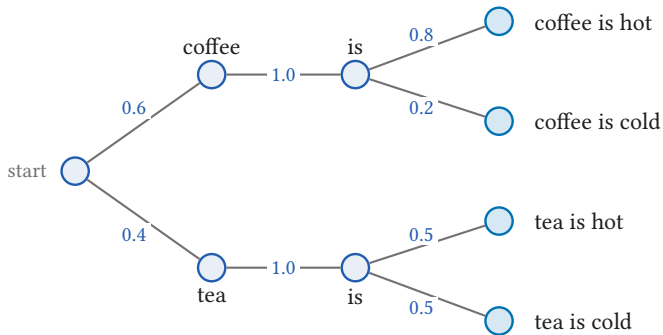
One table, every text it can write



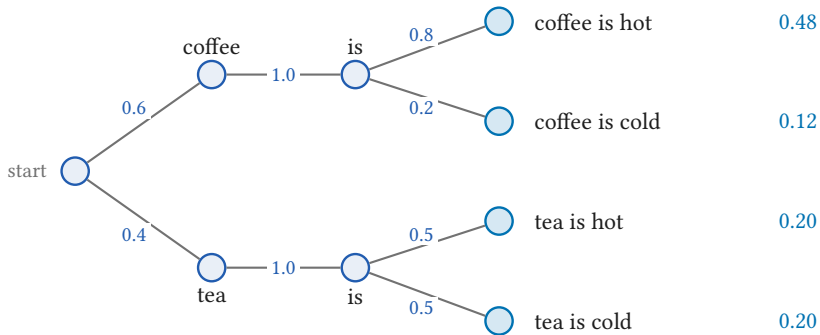
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One table, every text it can write

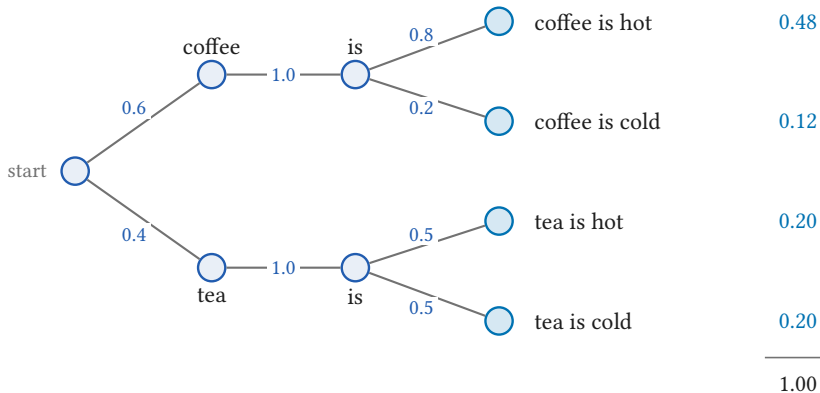


One table, every text it can write



Multiply along the path

One table, every text it can write

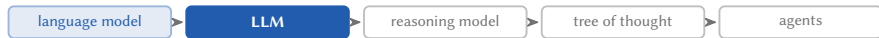




Section 3

Compressing the prefix

3



Real prefixes are long

Each report is a prefix, ending ...so tomorrow will be. The next word finishes the sentence.

It is cold and raining hard, ...

sunny	rainy	snowy
0	0.3	0.7

Real prefixes are long

Each report is a prefix, ending ...so tomorrow will be. The next word finishes the sentence.

It is cold and raining hard, ...

It is chilly and the air is dry, ...

	sunny	rainy	snowy
It is cold and raining hard, ...	0	0.3	0.7
It is chilly and the air is dry, ...	0.9	0	0.1

Real prefixes are long

Each report is a prefix, ending ...so tomorrow will be. The next word finishes the sentence.

It is cold and raining hard, ...

It is chilly and the air is dry, ...

It is warm and pouring with rain, ...

	sunny	rainy	snowy
It is cold and raining hard, ...	0	0.3	0.7
It is chilly and the air is dry, ...	0.9	0	0.1
It is warm and pouring with rain, ...	0.2	0.8	0

Real prefixes are long

Each report is a prefix, ending ...so tomorrow will be. The next word finishes the sentence.

It is cold and raining hard, ...

It is chilly and the air is dry, ...

It is warm and pouring with rain, ...

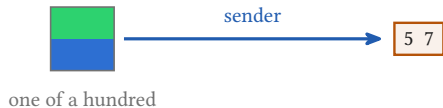
Every other report anyone could write, in any wording, at any length, ...

	sunny	rainy	snowy
It is cold and raining hard, ...	0	0.3	0.7
It is chilly and the air is dry, ...	0.9	0	0.1
It is warm and pouring with rain, ...	0.2	0.8	0
Every other report anyone could write, in any wording, at any length, ...	⋮	⋮	⋮

The toy table had six rows. This one has no last row.

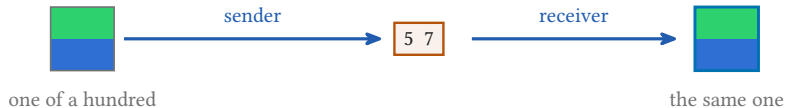
The yogurt cup game, again

Weekend 5



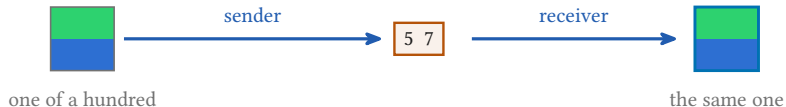
The yogurt cup game, again

Weekend 5



The yogurt cup game, again

Weekend 5

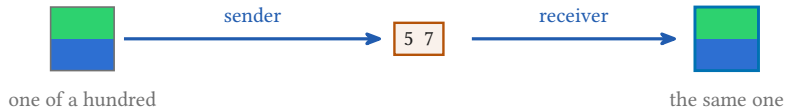


Today

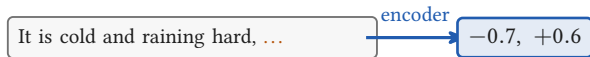
It is cold and raining hard, ...

The yogurt cup game, again

Weekend 5

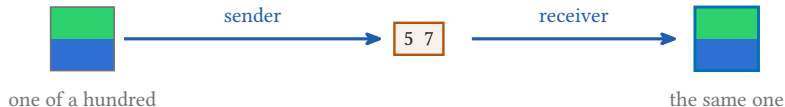


Today



The yogurt cup game, again

Weekend 5



Today



One network, not two: the layers that read the prefix and the head that turns the result into a row, trained together by predicting the next word

1)

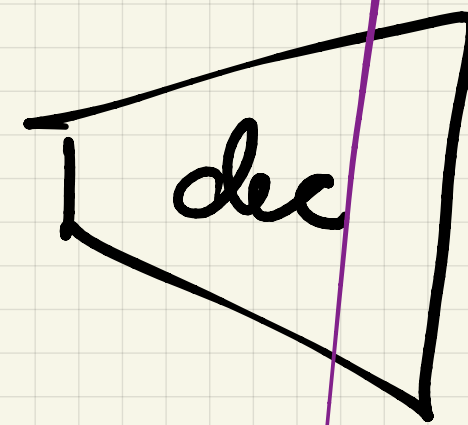
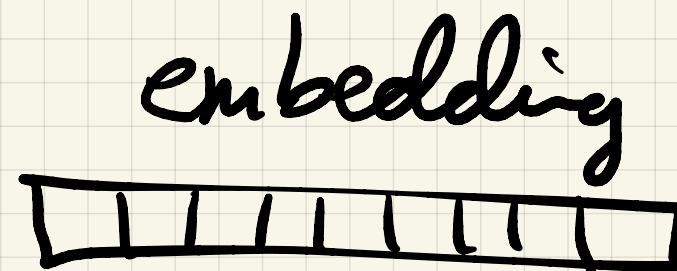
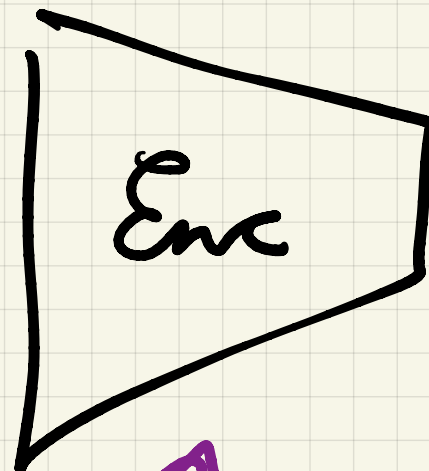
Two

NN's

play

yogurt cup

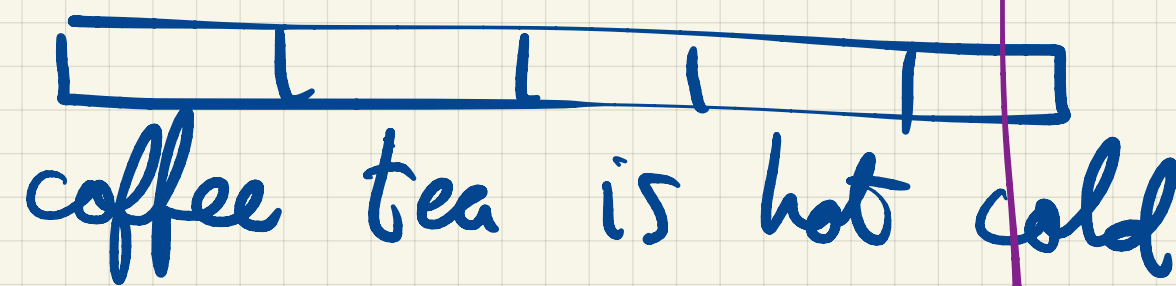
prefix



prefix

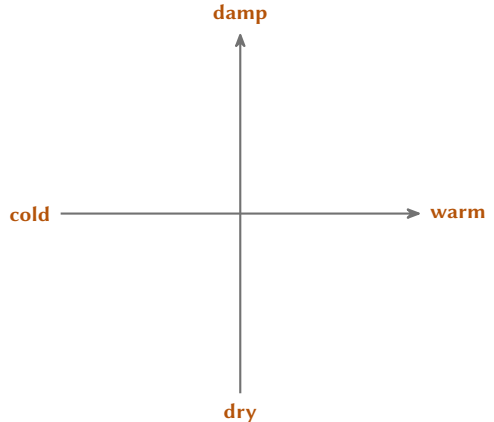
< 1TB

table row

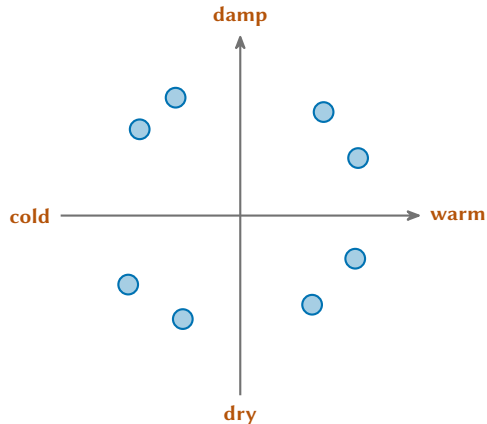


LLM

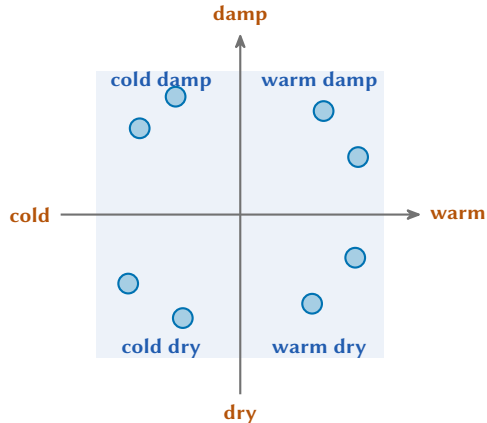
The prefix becomes two numbers



The prefix becomes two numbers



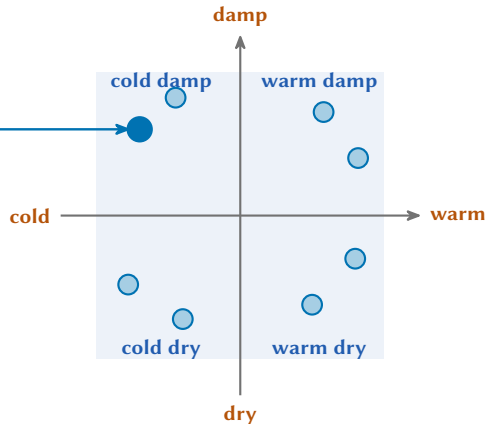
The prefix becomes two numbers



The prefix becomes two numbers

It is cold and raining hard, ...

Every report ends the same way: ...so tomorrow will be
A whole report in, two numbers out



Eight reports, four rows

Every report ends the same way: ...so tomorrow will be

- It is cold and raining hard, ...
- Freezing outside and the ground is wet, ...
- It is chilly and the air is dry, ...
- Icy morning and no rain at all, ...
- It is warm and pouring with rain, ...
- Hot outside and the ground is soaked, ...
- It is hot and completely dry, ...
- Mild morning and not a cloud anywhere, ...

sunny rainy snowy

Eight reports, four rows

Every report ends the same way: ...so tomorrow will be

- It is cold and raining hard, ...
- Freezing outside and the ground is wet, ...
- It is chilly and the air is dry, ...
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- It is warm and pouring with rain, ...
- Hot outside and the ground is soaked, ...
- It is hot and completely dry, ...
- Mild morning and not a cloud anywhere, ...

sunny rainy snowy

0	0.3	0.7
0	0.3	0.7

cold damp

Eight reports, four rows

Every report ends the same way: ...so tomorrow will be

- It is cold and raining hard, ...
- Freezing outside and the ground is wet, ...
- It is chilly and the air is dry, ...
- Icy morning and no rain at all, ...
- It is warm and pouring with rain, ...
- Hot outside and the ground is soaked, ...
- It is hot and completely dry, ...
- Mild morning and not a cloud anywhere, ...

sunny rainy snowy

0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	

Eight reports, four rows

Every report ends the same way: ...so tomorrow will be

- It is cold and raining hard, ...
- Freezing outside and the ground is wet, ...
- It is chilly and the air is dry, ...
- Icy morning and no rain at all, ...
- It is warm and pouring with rain, ...
- Hot outside and the ground is soaked, ...
- It is hot and completely dry, ...
- Mild morning and not a cloud anywhere, ...

sunny rainy snowy

0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	
0.2	0.8	0	warm damp
0.2	0.8	0	

Eight reports, four rows

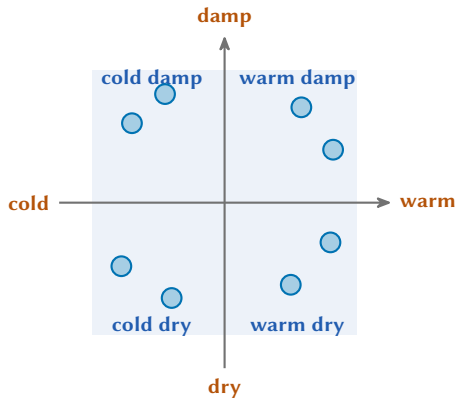
Every report ends the same way: ...so tomorrow will be

- It is cold and raining hard, ...
- Freezing outside and the ground is wet, ...
- It is chilly and the air is dry, ...
- Icy morning and no rain at all, ...
- It is warm and pouring with rain, ...
- Hot outside and the ground is soaked, ...
- It is hot and completely dry, ...
- Mild morning and not a cloud anywhere, ...

sunny rainy snowy

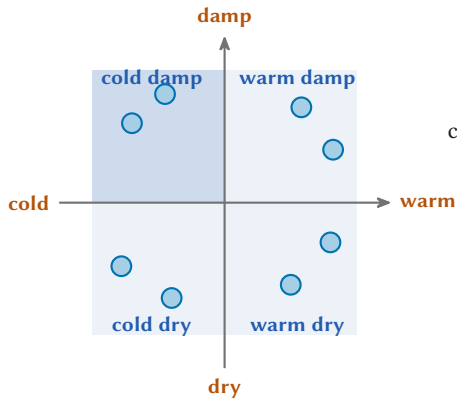
0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	
0.2	0.8	0	warm damp
0.2	0.8	0	
0.9	0.1	0	warm dry
0.9	0.1	0	

One row for each region



sunny rainy snowy

One row for each region

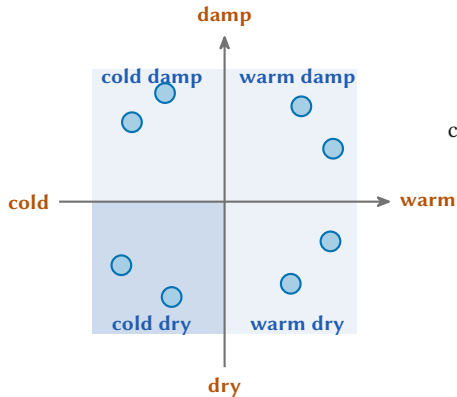


cold damp

sunny rainy snowy

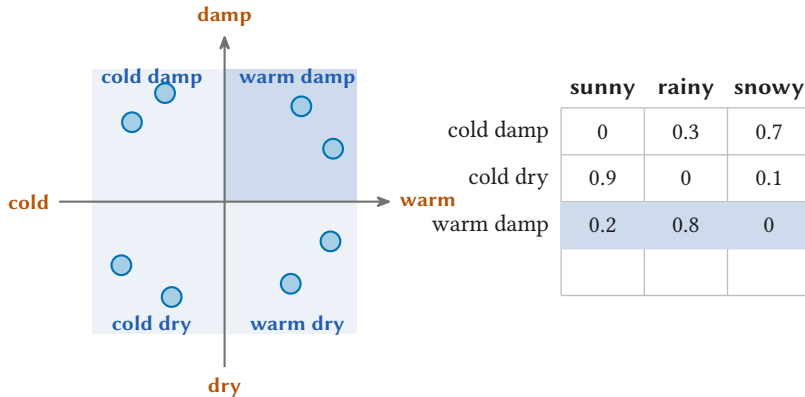
sunny	rainy	snowy
0	0.3	0.7

One row for each region

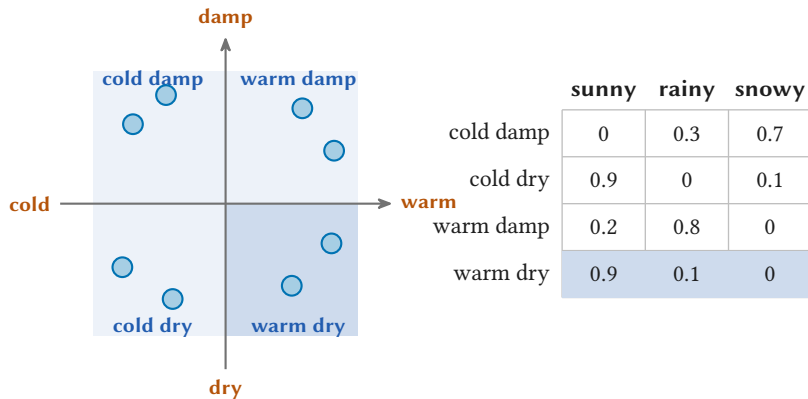


	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1

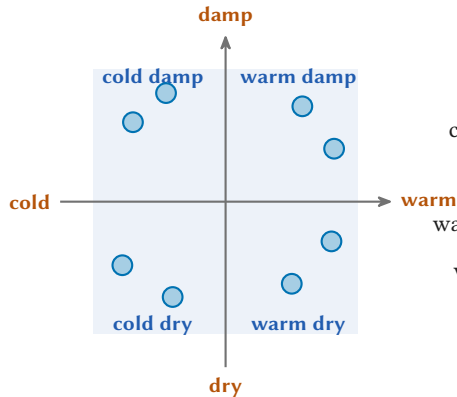
One row for each region



One row for each region



One row for each region



	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1
warm damp	0.2	0.8	0
warm dry	0.9	0.1	0

Cold alone gives 0.7
against 0.1.
Damp alone gives 0.7
against 0.0.
Only both give snowy.



Section 4

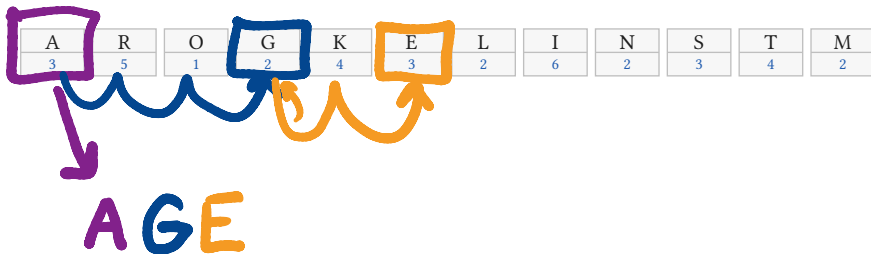
Why writing helps

4



The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.



The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

A	R	O	G	K	E	L	I	N	S	T	M
3	5	1	2	4	3	2	6	2	3	4	2

straight off the strip

A R O G K E L I N S T M



The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

A	R	O	G	K	E	L	I	N	S	T	M
3	5	1	2	4	3	2	6	2	3	4	2

straight off the strip A R O G K E L I N S T M → ?

the recipe spells A

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.



straight off the strip A R O G K E L I N S T M → ?

the recipe spells

A G E

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.



straight off the strip A R O G K E L I N S T M → ?

the recipe spells

A G E N T

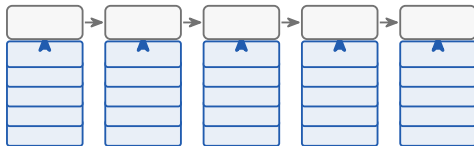
What the extra words buy

- Every hop is easy. There are simply too many in a row for one pass.



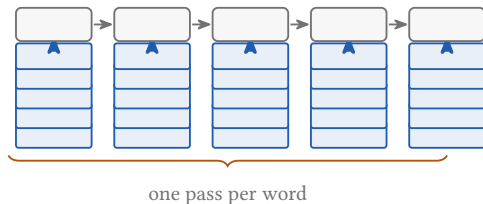
What the extra words buy

- Every hop is easy. There are simply too many in a row for one pass.
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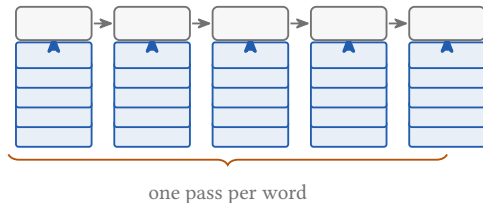
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What the extra words buy

- Every hop is easy. There are simply too many in a row for one pass.
- Each written word is the memory the next pass reads.
- The depth is fixed. The number of passes is not.
- With no intermediate words a transformer is confined to one shallow circuit class.



Merrill and Sabharwal. The Expressive Power of Transformers with Chain of Thought. ICLR 2024.
Feng, Zhang, Gu et al. Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective. NeurIPS 2023.



Section 5

Training the model to reason

5

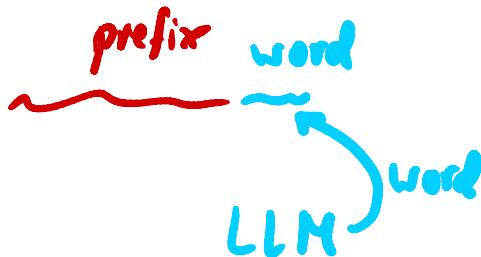


The chain of thought as a decision problem

RL

What is 17 x 23?

s_0



States

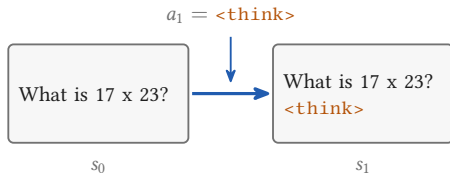
Actions

TD

Reward

state the prompt plus everything written so far, so the state grows by one token at every step

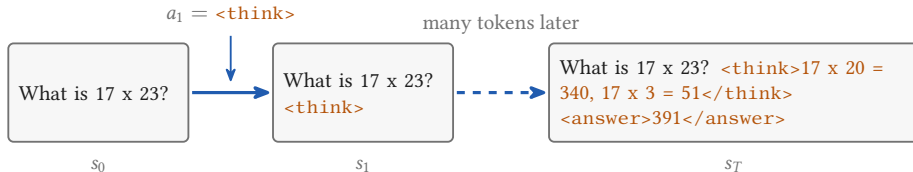
The chain of thought as a decision problem



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action the next token, chosen by the model, which is the policy

The chain of thought as a decision problem

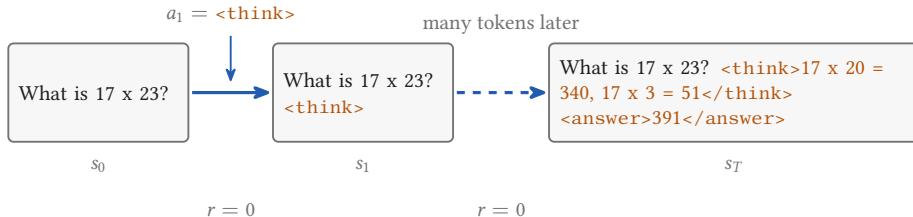


state the prompt plus everything written so far, so the state grows by one token at every step

action the next token, chosen by the model, which is the policy

format reasoning inside <think> tags, answer inside <answer> tags, so a rule can find it

The chain of thought as a decision problem



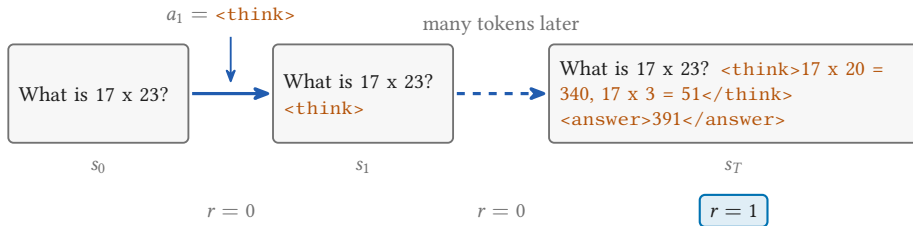
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format reasoning inside `<think>` tags, answer inside `<answer>` tags, so a rule can find it

reward zero until the answer is written, then 1 if it is right and 0 if it is wrong

The chain of thought as a decision problem



state the prompt plus everything written so far, so the state grows by one token at every step

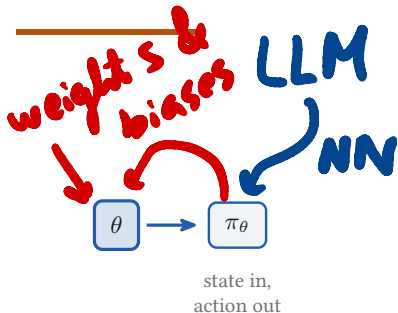
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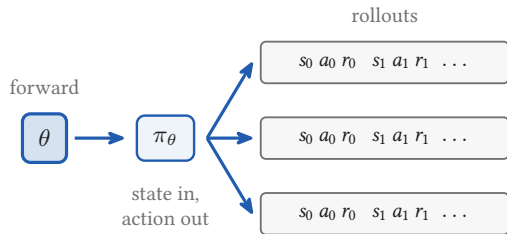
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Guo, Yang, Zhang et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature 645, 633 to 638, 2025.

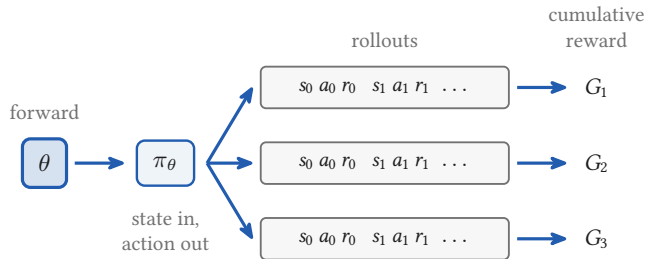
How reinforcement learning trains a policy



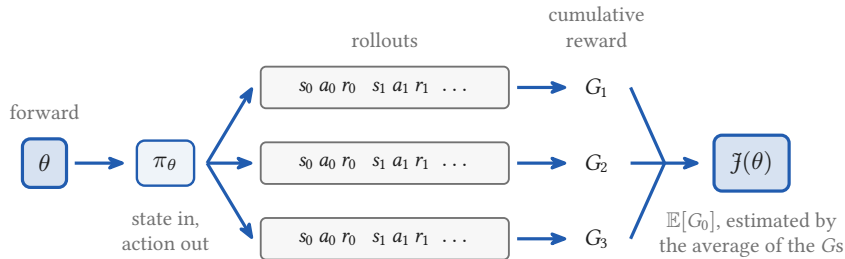
How reinforcement learning trains a policy



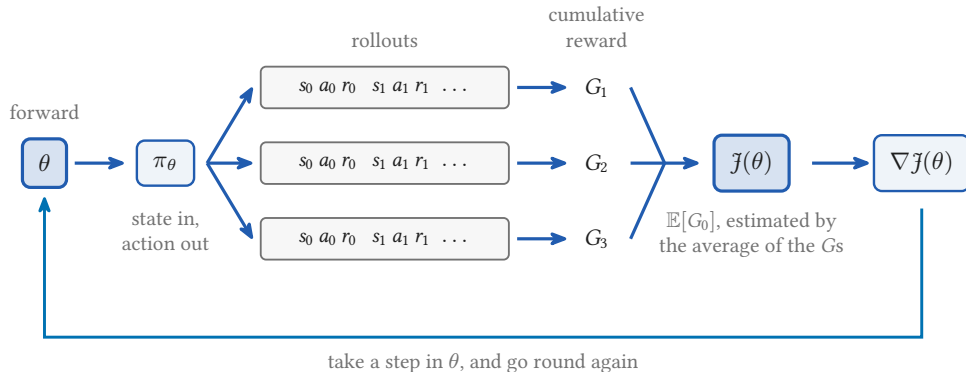
How reinforcement learning trains a policy



How reinforcement learning trains a policy



How reinforcement learning trains a policy



Weekend 4, the same loop with a point in place of a policy: [the grasshopper on a hill it cannot see](#)

The gradient, term by term

$$\nabla_{\theta} \mathcal{J}(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t \Psi_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

The expectation is over episodes generated by π_{θ} ; in practice, the average over the rollouts just collected.

The gradient, term by term

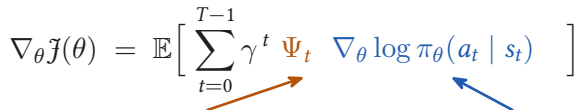
$$\nabla_{\theta} \tilde{J}(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t \Psi_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$


the score

which way to nudge θ so that a_t
becomes more likely at s_t

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The gradient, term by term

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the multiplier

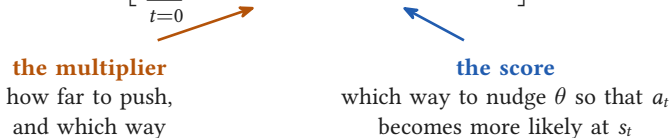
how far to push,
and which way

the score

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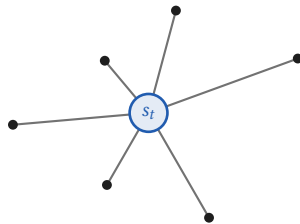
which way to nudge θ so that a_t
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The expectation is over episodes generated by π_{θ} ; in practice, the average over the rollouts just collected.

The logarithm makes the nudge relative: a rare action that pays off moves further than one the policy already takes.

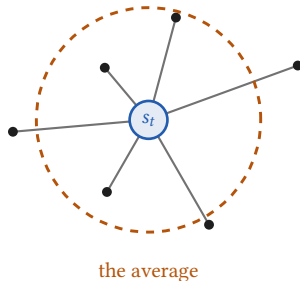
How much to believe the direction

- One spoke per action available at s_t , as long as that action turned out to be worth.



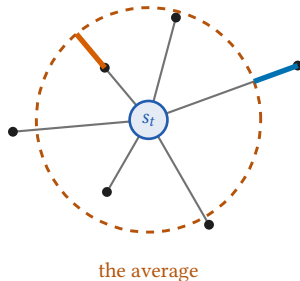
How much to believe the direction

- One spoke per action available at s_t , as long as that action turned out to be worth.
- The circle sits at their average: what the state itself is worth.



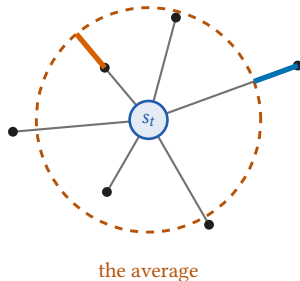
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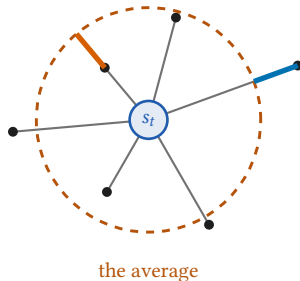
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How much to believe the direction

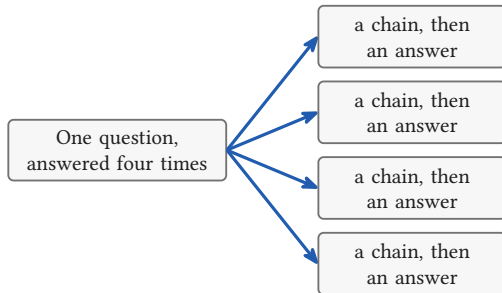
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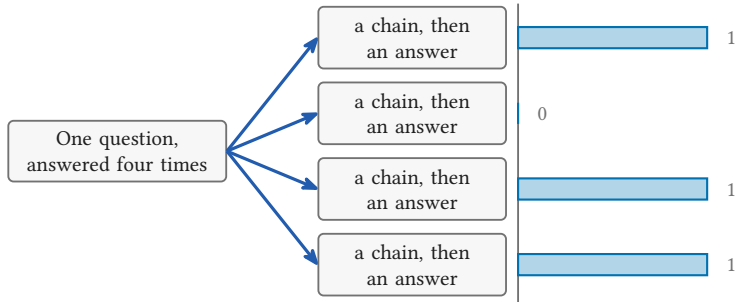
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For a model that reasons, the **actor** is the language model itself. And the critic?

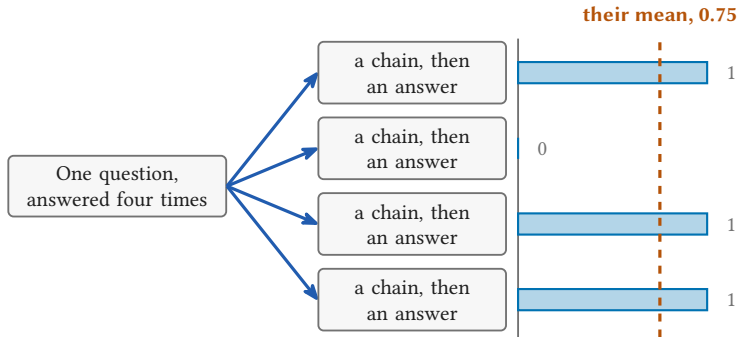
The group is the baseline



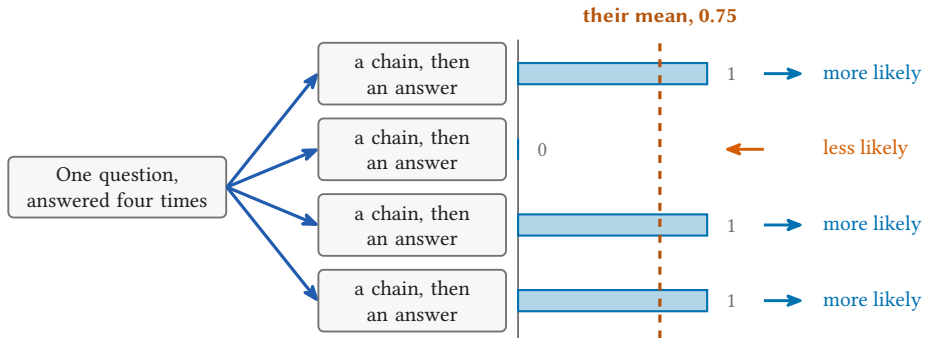
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The group is the baseline



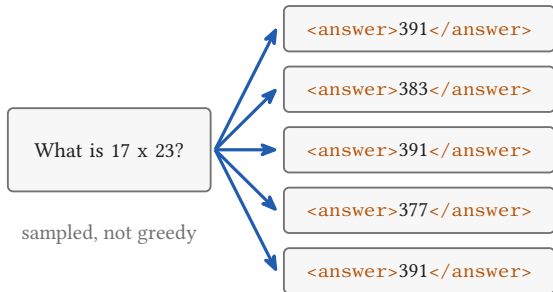
Guo, Yang, Zhang et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature 645, 633 to 638, 2025.

Asking the same question several times

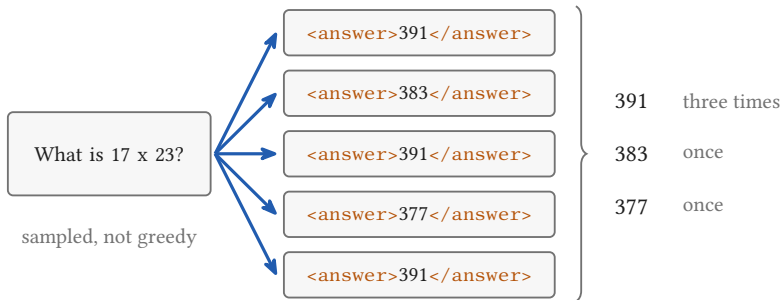
What is 17×23 ?

sampled, not greedy

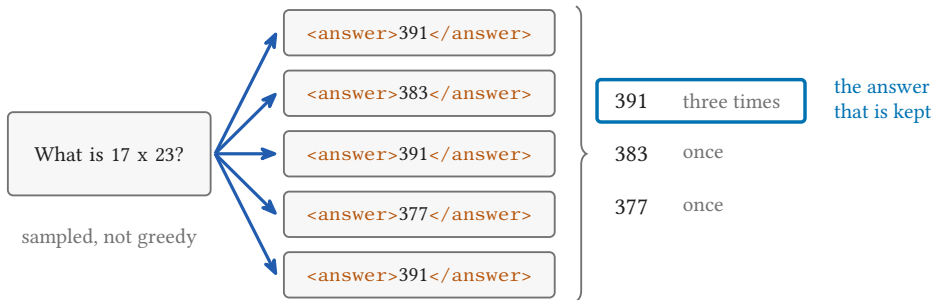
Asking the same question several times



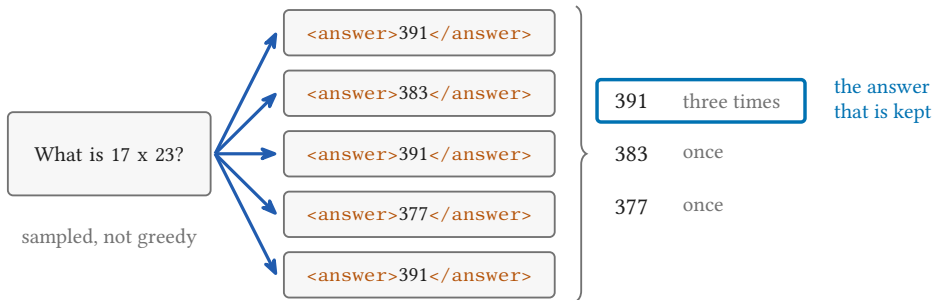
Asking the same question several times



Asking the same question several times



Asking the same question several times



- No weight changes. This is a way of decoding, not a way of training.
- One vote costs five whole answers, so voting and not voting are not the same measurement.

Wang, Wei, Schuurmans et al. Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023.

Reasoning from pure reinforcement learning

- DeepSeek-R1-Zero starts from the base model:
no supervised fine-tuning, no worked examples
of reasoning.

Reasoning from pure reinforcement learning

AIME 2024, pass@1

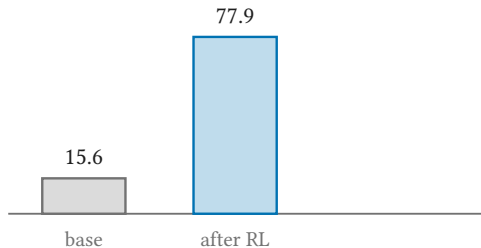
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Reasoning from pure reinforcement learning

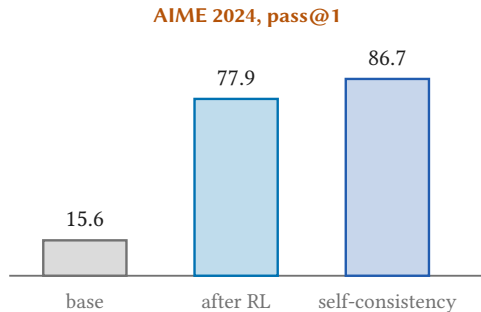
AIME 2024, pass@1

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Reasoning from pure reinforcement learning

- DeepSeek-R1-Zero starts from the base model: no supervised fine-tuning, no worked examples of reasoning.
- Longer chains and self-checking appeared on their own.



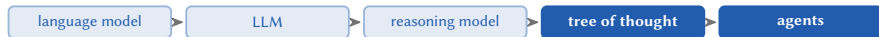
Guo, Yang, Zhang et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature 645, 633 to 638, 2025.

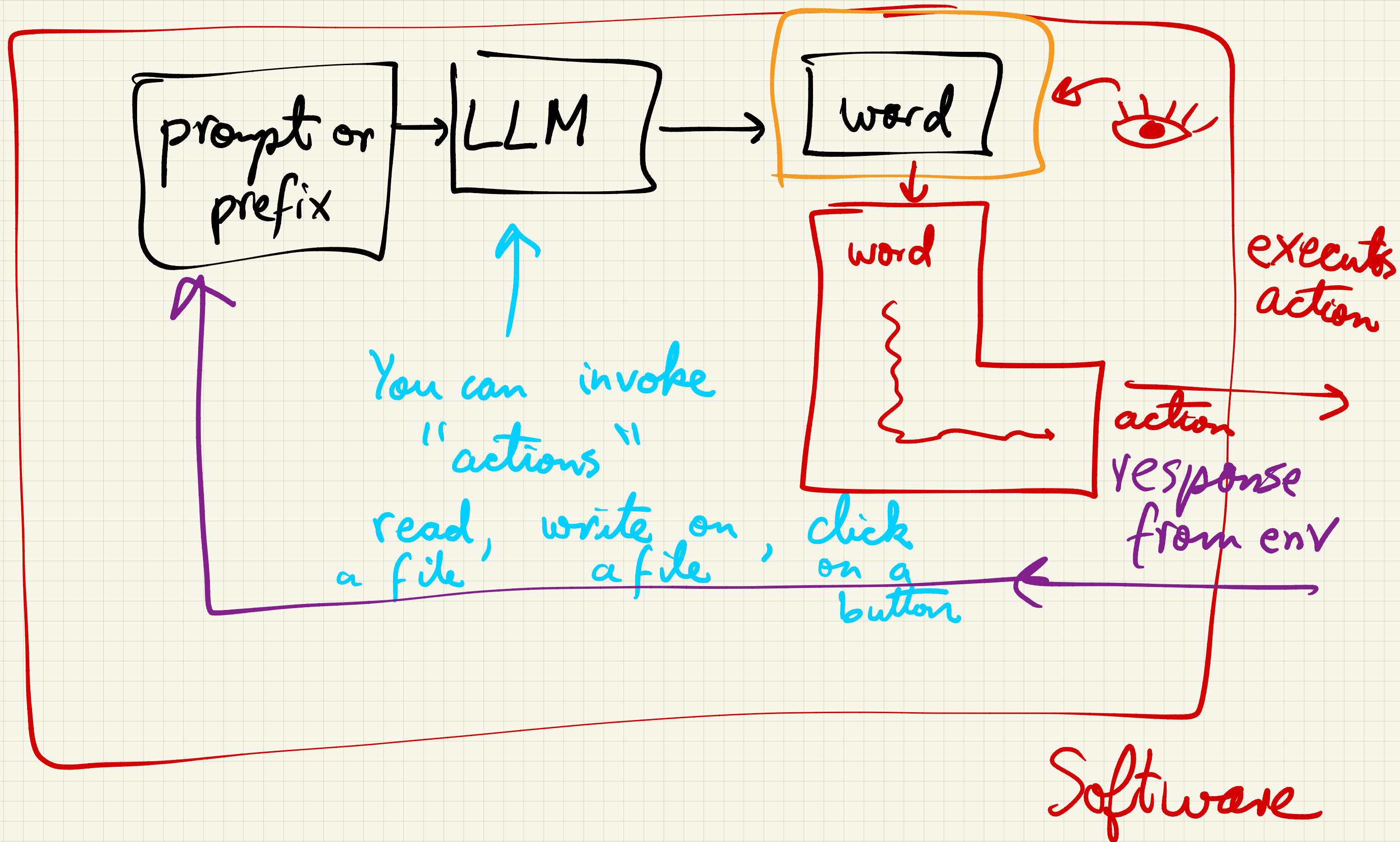


Section 6

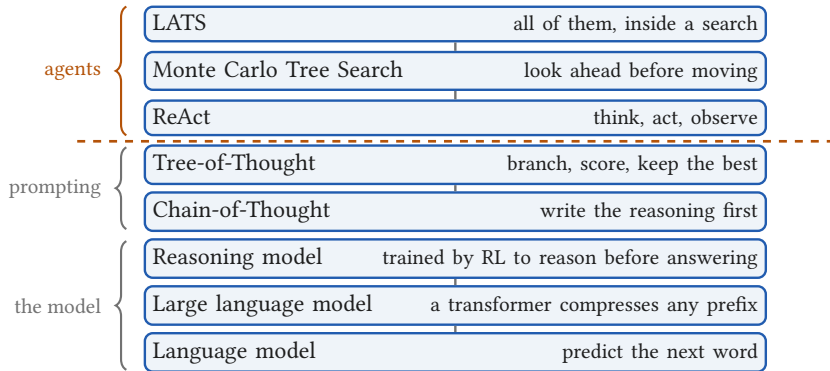
From prompting to agents

6

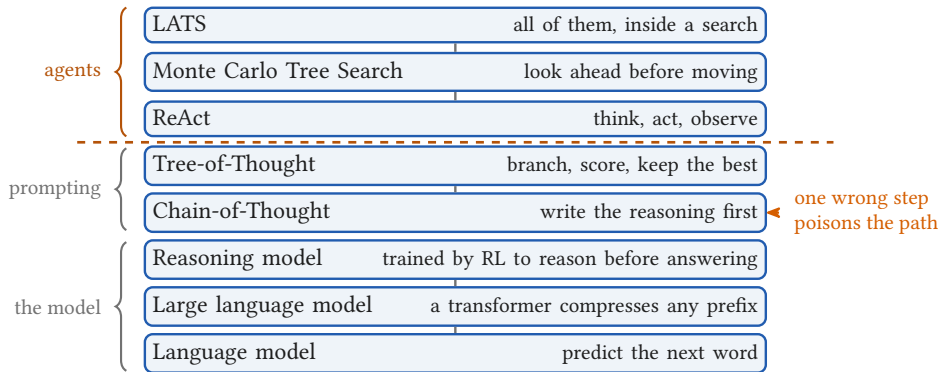




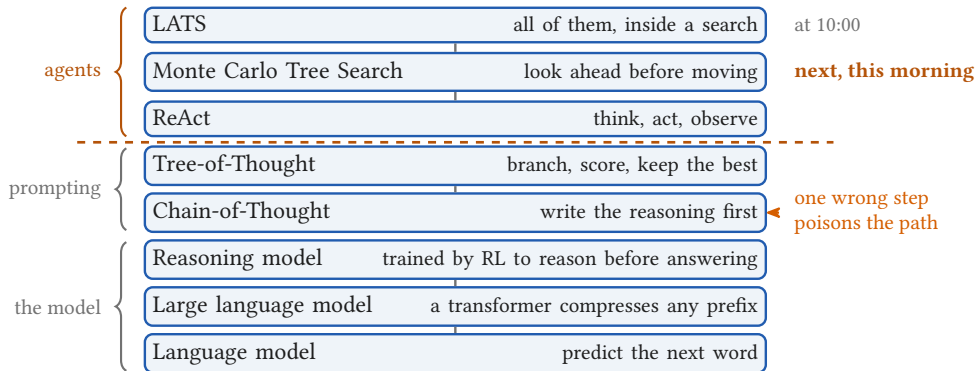
From a table to an agent



From a table to an agent



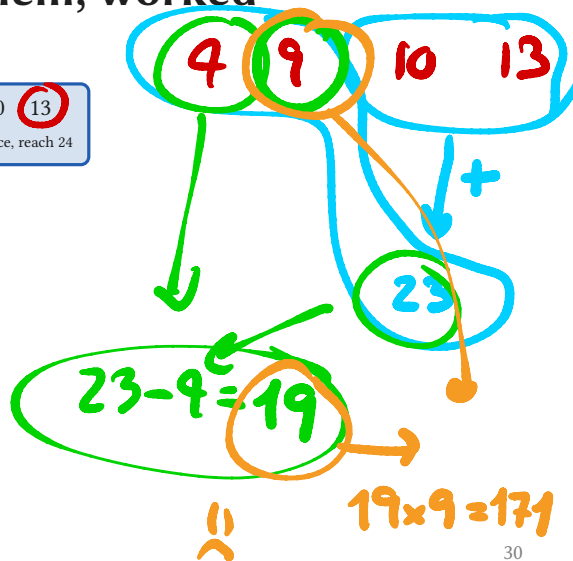
From a table to an agent



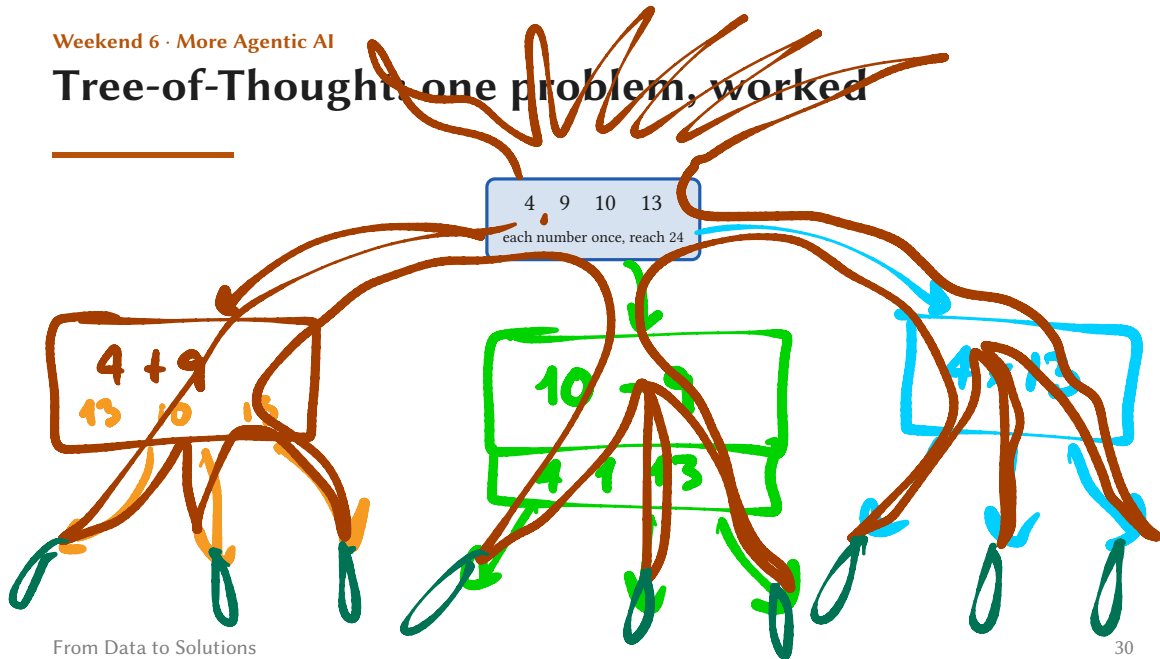
Wei et al. 2022 (Chain-of-Thought), Yao et al. 2023 (Tree of Thoughts, ReAct), Guo et al. 2025 (DeepSeek-R1), Zhou et al. 2024 (LATS).

Tree-of-Thought: one problem, worked

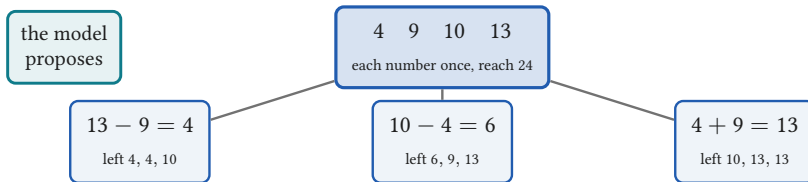
4 9 10 13
each number once, reach 24



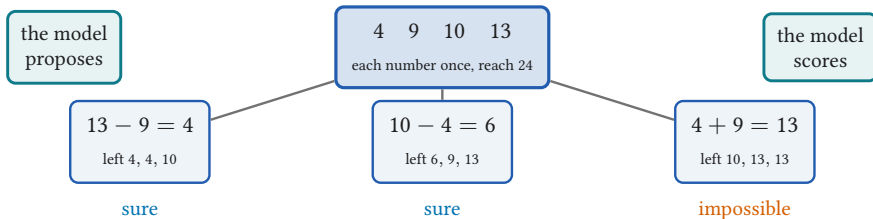
Tree-of-Thought one problem, worked



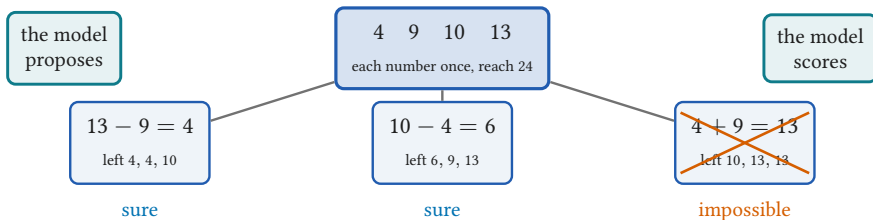
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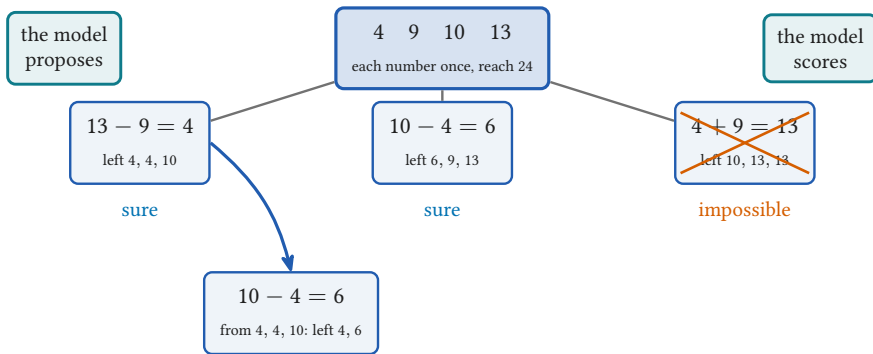
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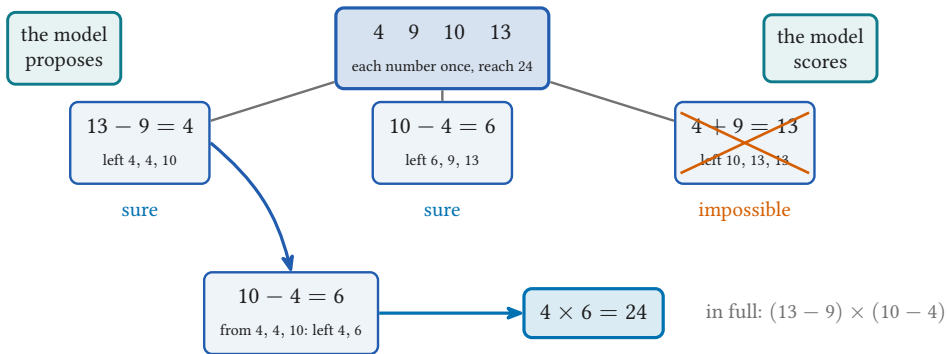
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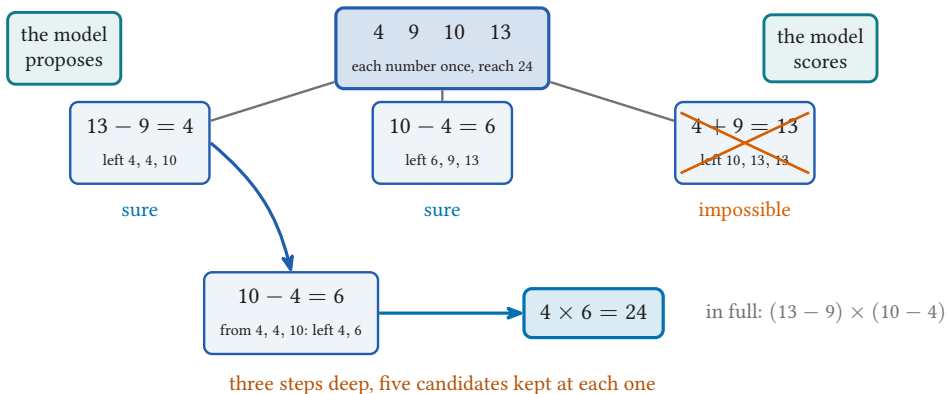
Tree-of-Thought: one problem, worked



Tree-of-Thought: one problem, worked



Tree-of-Thought: one problem, worked



Tree-of-Thought: what the search buys

Game of 24, GPT-4

chain of thought  4 %

Tree-of-Thought: what the search buys

Game of 24, GPT-4

chain of thought  4 %

Tree-of-Thought  74 %

Tree-of-Thought: what the search buys

Game of 24, GPT-4

chain of thought  4 %

Tree-of-Thought  74 %

5x5 crosswords, words filled correctly

Tree-of-Thought  20 %

Tree-of-Thought: what the search buys

Game of 24, GPT-4

chain of thought 4 %

Tree-of-Thought 74 %

5x5 crosswords, words filled correctly

Tree-of-Thought 20 %

without pruning 5 %

without backtracking 5 %

take away the pruning, or the backtracking,
and almost all of it goes

Tree-of-Thought: what the search buys

Game of 24, GPT-4

chain of thought 4 %

Tree-of-Thought 74 %

5x5 crosswords, words filled correctly

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take away the pruning, or the backtracking,
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The authors search the tree breadth first and depth first, and call those [relatively simple](#). They leave A* and Monte Carlo Tree Search to future work. That is the rest of this morning.

Backup

frames kept for questions

A regression problem in the prompt

the prompt

$1 \rightarrow 3$

$2 \rightarrow 5$

$3 \rightarrow 7$

A regression problem in the prompt

the prompt

$1 \rightarrow 3$

$2 \rightarrow 5$

$3 \rightarrow 7$

$4 \rightarrow ?$

A regression problem in the prompt

the prompt

$1 \rightarrow 3$

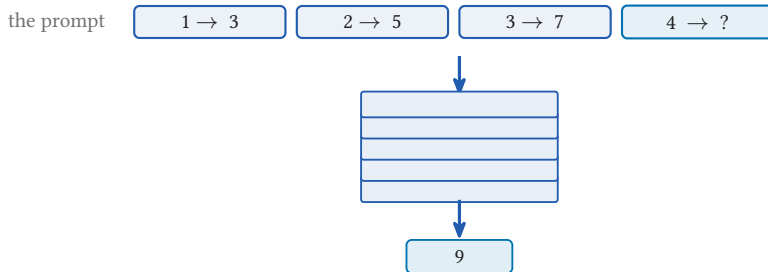
$2 \rightarrow 5$

$3 \rightarrow 7$

$4 \rightarrow ?$



A regression problem in the prompt



One pass. No weight changed, and the pairs exist only in the prompt

The layers can do the fitting



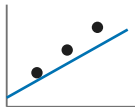
after layer 1

- By explicit construction, the layers can carry out gradient descent.

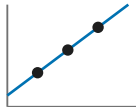
The layers can do the fitting



after layer 1



after layer 2



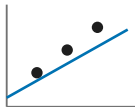
after layer 3

- By explicit construction, the layers can carry out gradient descent.
- So one forward pass can solve a basic regression problem, shown for linear models.

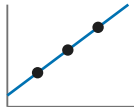
The layers can do the fitting



after layer 1



after layer 2



after layer 3

- By explicit construction, the layers can carry out gradient descent.
- So one forward pass can solve a basic regression problem, shown for linear models.

Åkyurek, Schuurmans, Andreas et al. What learning algorithm is in-context learning? Investigations with linear models. ICLR 2023.
von Oswald, Niklasson, Randazzo et al. Transformers Learn In-Context by Gradient Descent. ICML 2023.